



MINING DEALER BEST PRACTICE

Oil Patch Analysis During Engine Dyno Test

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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December 2024
4.01-241220-1

1.0 Introduction

To seek continuous improvements in quality, processes and procedures, that Matco perform daily at the CRC and to ensure consistent and reliable component performance is achieved in the field. Matco designed and custom built a unique dual oil patch sample tooling, for each dyno cell which is situated in the control room of the dyno cells, eliminating the need for the cell operator to enter the dyno cell, multiple times during any engine testing, to remove/change the sample patch for inspection and taking multiple patch samples at three five minute intervals, will assist in identifying any issues that potentially could result in considerable internal engine damage, if no patch sample testing was performed during the dyno test.

2.0 Best Practice Description

Using this patch sampling, multiple times within the environment of the control room, while the engine is operating at the various speeds and load points, provides improved management of each patch sample.

For storage purposes and history of each engine patch test. Matco developed a Patch Test Service Report (SR) format that is used to place the patch on, including at what time interval the patch was inspected at, as seen in picture 1.

		PATCH TEST DYNAMOMETER REPORT		CODE
Quality Management Systems ISO 9001				FO-CRC-09 R0
ORDER NUMBER :		DATE:		
ENGINE NUMBER:		CLIENT:		
SERIAL NUMBER:		REPAIR:		
		0-5 MINUTES		1/2 LOAD
		RPM		RPM
		LOAD HP		LOAD HP
		0.00%		50.00%
		OIL TEMP (°F)		OIL TEMP (°F)
		5-10 MINUTES		FULL LOAD
		RPM		RPM
		LOAD HP		LOAD HP
		0.00%		100.00%
		OIL TEMP (°F)		OIL TEMP (°F)
		10-15 MINUTOS		TRQ MAXIMO
		RPM		RPM
		LOAD HP		LOAD HP
		25.00%		100.00%
		OIL TEMP (°F)		OIL TEMP (°F)

TECHNICIAN: _____

SUPERVISOR: _____

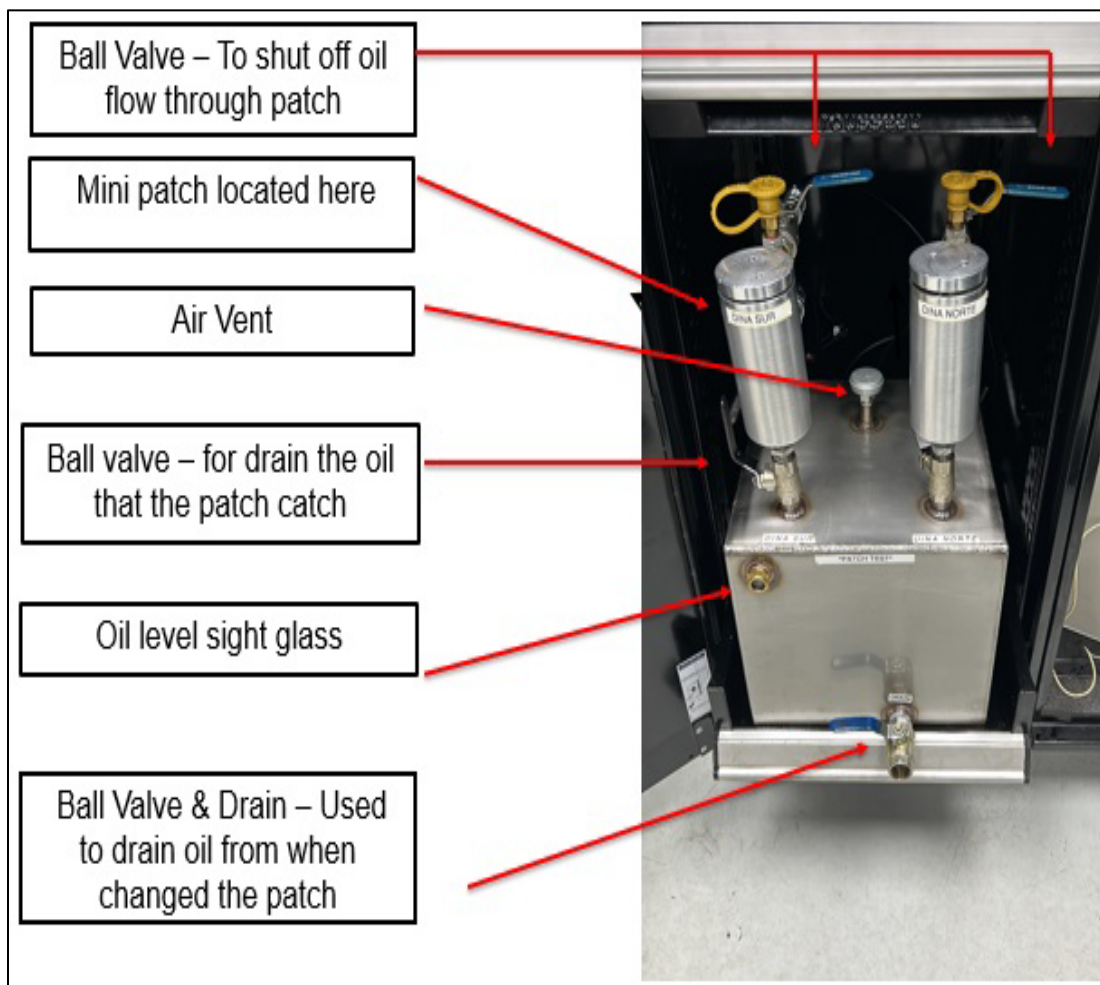
Picture 1: Format used to record the oil test samples.

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Oil Patch Analysis During Engine Dyno Test	DATE 12/20/2024	CHG NO 00	NUMBER 4.01-241220-1

During the dyno test, pressure oil is directed from the engine, prior to the engine filter, through one hose that connects the tooling above the sample patch, this patch will then collect any debris for analysis and the oil passing through the patch, is then returned to the engine.

The purpose of each patch test at the various intervals, is to ensure we have very clean samples with no debris and if for any reason, metallic debris were identified, then it would result in an immediate engine shut down, to investigate root causes.

As shown in picture 2, this is the individual patch tooling, used for both dyno cells which is in the control center. There are two inlet and two return lines one for each dyno cell. When the inlet valve is closed to remove the patch, the lower valve is open, to permit oil drainage into the tank.



Picture 2: This are the parts used in the patch test system.

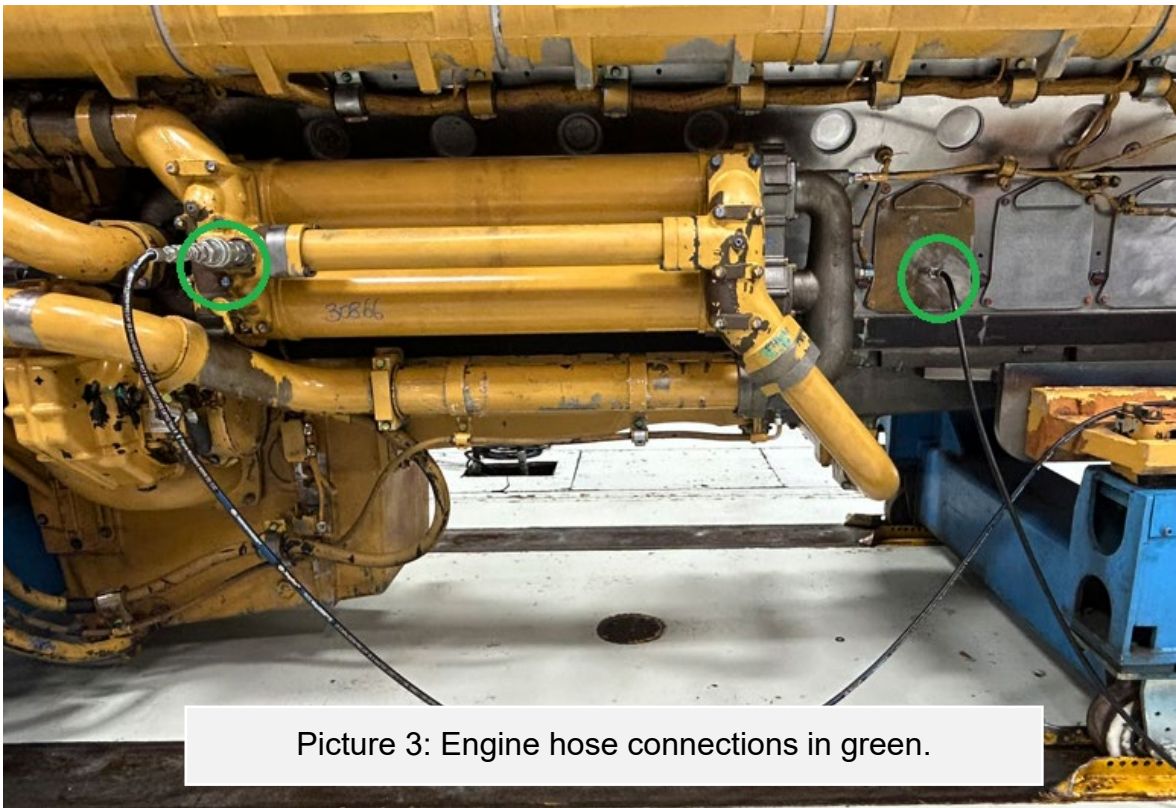
3.0 Implementation Steps

With the engine inside of the Dyno Cell one hose is connected, at the oil cooler adapter to the interface monitor (communication control between the engine and monitor panel) as shown in

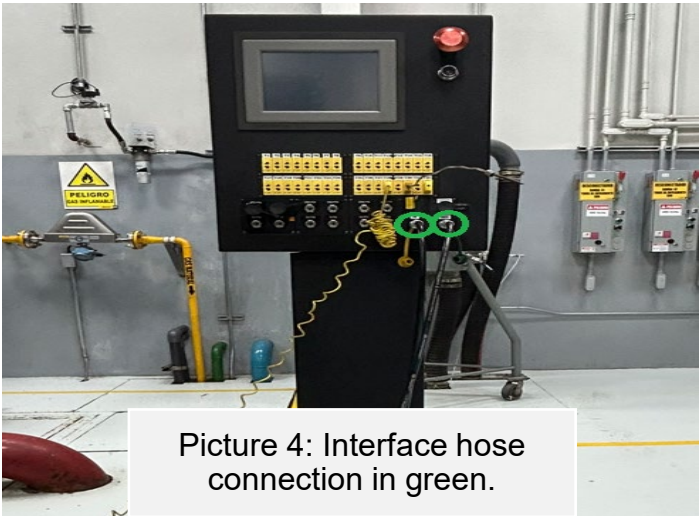
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picture 4. The hose from the monitoring panel goes directly into the enclosed operator booth where the patch test tooling is located.

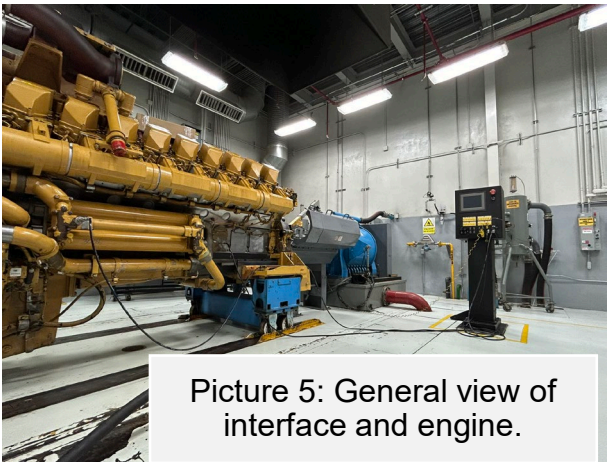
Shown in picture 3, 4 and 5 is an example of a C175-20 engine connection.



Picture 3: Engine hose connections in green.



Picture 4: Interface hose connection in green.



Picture 5: General view of interface and engine.

Next, we place the screen and a clean patch in the canister and install the cap, as shown in picture 6, and when the engine is started, pressure oil from the engine prior to the oil filter will start to flow through the patch and after five minutes the inlet control valve is then closed to the remove the first patch while the test is still running. Each patch is then replaced by a new one and the valve is

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opened again to allow oil flow. This process will be repeated several times during the engine test. Picture 7 is an example of the patch samples at the various times.



Picture 6

FO-CRC-99 Rep. de Patch test

MATCO CAT		REPORTE PATCH TEST DINAMOMETRO		CLAVE
Sistema de monitoreo de la Central 153 0001				FO-CRC-09 R0
NUMERO DE ORDEN:	CR-82-08115	FECHA:	14-mayo-24	
MODELO MOTOR:	C17509	CLIENTE:	OPERADORA DE MINAS	
NUMERO DE SERIE:	ZK3112	REPARACION:	GENERAL	
0-5 MINUTOS		5/2 LOAD		
RPM:	700	RPM:	1833	
LOAD HP:	6.00%	LOAD HP:	10.80%	
OIL TEMP (°F):	95	OIL TEMP (°F):	203	
5-10 MINUTOS		FULL LOAD		
RPM:	1543	RPM:	1790	
LOAD HP:	9.00%	LOAD HP:	17.00%	
OIL TEMP (°F):	117	OIL TEMP (°F):	205	
10-15 MINUTOS		TRQ MAXIMO		
RPM:	1506	RPM:	1360	
LOAD HP:	25.00%	LOAD HP:	150.00%	
OIL TEMP (°F):	184	OIL TEMP (°F):	206	
TECNICO: Carlos Martinez		SUPERVISOR: Jose Alcaraz - 14/05/2024		
Página 1				

Picture 7

During and after the dyno test is concluded, the patches are then labeled by sequence and analyzed by the technician. The results are then recorded by pictures and on the paper format for filing purposes.

4.0 Benefits

- Multiple filter samples can be taken throughout the programmed dyno test.
- Cost saving by eliminating filter replacement after the dyno test
- No filters are removed in dyno cell
- Rapid detection of any debris.
- Eliminates potential costly failures during the dyno test.

5.0 Resources Required

Fabricate Patch Tooling.

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6.0 Supporting Attachments / References

Patch Test Report (see Appendix)

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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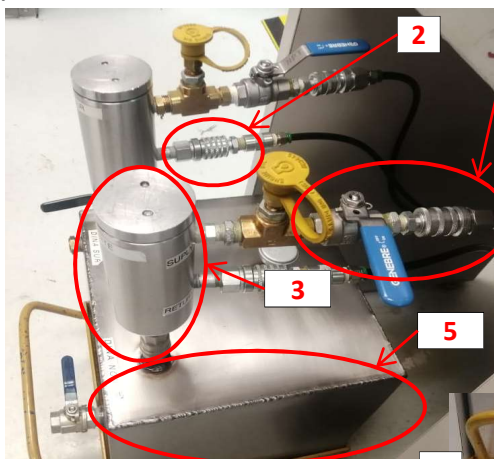
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		PROCESS SHEET														
		Dynamometer Patch Test														
		Area	Section													
Folio:	IT-CRC-07	Dynamometer	Filter Test													
Supervisor:		Page:	Version:	Date:												
Dynamometer Technician		1 de 3	00	24/11/2021												
1) Filter module    A) Patch test module components 1) Oil inlet 2) Oil Return 3) Chamber / Patch section 1E2500D 4) Oil purge to tank 5) Oil tank 6) Oil tank drain 2) Patch installation   A) Open the patch chamber and instal the mini patch number 1E2500D in the lower part of the chamber		3) Oil filter test process Nota: Previously to the next step, make sure to follow the process shown in IT-CRC-07 A) Test module valve verification 1) Inlet valve (1) must be open. 2) Return valve (2) must be open. 3) Purge valve to oil tank (4) must be closed. 4) Patch chamber latches must be up securing the conection. 5) Oil drain valve from tank must be closed. 4) Patch change 1E2500D <table border="1"> <thead> <tr> <th>Patch</th> <th>Process</th> <th>Time</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>Warm up</td> <td>0 - 10 min</td> </tr> <tr> <td>2</td> <td>Warm up</td> <td>10 - 15 min</td> </tr> <tr> <td>3</td> <td>Final Load</td> <td>Test Duration</td> </tr> </tbody> </table> A) Test module valve verification 1) Inlet valve (1) must be closed. 2) Return valve (2) must be closed. 3) Purge valve to oil tank (4) must be open. 4) Oil tank drain valve must be open (6) 5) Leave the drain valve open for 5 minutes to purge all the oil. 6) Change the test patch. 7) Repeat the process as necessary. Note: After two complete engine tests, the oil tank must be empty and cleaned.			Patch	Process	Time	1	Warm up	0 - 10 min	2	Warm up	10 - 15 min	3	Final Load	Test Duration
Patch	Process	Time														
1	Warm up	0 - 10 min														
2	Warm up	10 - 15 min														
3	Final Load	Test Duration														
Quality and safety consideration: 1. Make sure there is no visible leakage in the connections and that the oil tank is not completely full. 2. When the patch is placed on the chamber, verify there is no visible contamination. 3. When in doubt, ask the supervisor before acting. 4. Wear your personal protective equipment at all times.			Review	Date												
			00													
			01													
			02													
			03													
			04													

		PROCESS SHEET					
		Dynamometer Patch Test					
		Area	Section				
Folio:	IT-CRC-07	Dynanometer	Filter Test				
Supervisor:		Page:	Version:	Date:			
Dynamometer Technician		2 de 3	00	24/11/2021			
1) Identify the engine model and fixtures needed.  (A)  (B)  (C)  (D) A) Engines 3500 and C175 have an oil check on one of the covers and a quick connect adaptor (B) B) Engines C32 and C27 will use the oil tank inlet adaptor (C) and the adaptor (B) installed in the oil filter base of the engine (D)		Engine Model C32 y C27   (G) (H) A) The inlet hose for the patch test must be before the oil filter. SOS sample connection in the filter base (H) B) The return hose of the patch test must be connected in a no pressure zone, oil inlet tube (G).					
Engine Model 3500 and C175		2) Fixtures and adaptors intallation   (E) (F) A) The inlet hose for the patch test must be connected before the oil filter of the engine. Hose connected to the heat exchanger (E) B) The return hose of the patch test must be connected in a no pressure zone. Hose connected in the crankshaft cover(F).			4) Fixtures cleaning and storage A) Follow the contamination control guidelines to properly clean and store the fixtures to make sure there is no external contamination that can disturb the test. Note: After the patch test is done, clean the fixtures with solvent to store them in the assigned cabinet (VIDMAR).		
Quality and safety consideration:		Review	Date				
1. Make sure there is no visible leakage in the connections and that the oil tank is not completely full.		00					
2. When the patch is placed on the chamber, verify there is no visible contamination.		01					
3. Verify that all the connections and fixtures are properly tight.		02					
3. When in doubt, ask the supervisor before acting.		03					
4. Wear your personal protective equipment at all times.		04					

		PROCESS SHEET	
		Dynamometer Patch Test	
		Area	Section
Folio:	IT-CRC-07	Dynanometer	Filter Test
Supervisor:		Page:	Version:
Dynamometer Technician		3 de 3	00
		Date:	
		24/11/2021	

1) Engine Test Report Document

A) The Date, Work Oder, Engine Serial Number, Engine Model, and client will be recorded on the Document.

MATCO CAT		PATCH TEST DYNAMOMETER REPORT		CODE
				FO-CRC-09 R0
ORDER NUMBER :	DATE:			
ENGINE NUMBER:	CLIENT:			
SERIAL NUMBER:	REPAIR:			
		0-5 MINUTES	1/2 LOAD	
		RPM	RPM	
		LOAD HP	LOAD HP	
		0.00%	50.00%	
		OIL TEMP (°F)	OIL TEMP (°F)	
		5-10 MINUTES	FULL LOAD	
		RPM	RPM	
		LOAD HP	LOAD HP	
		0.00%	100.00%	
		OIL TEMP (°F)	OIL TEMP (°F)	
		10-15 MINUTES	MAX TORQ	
		RPM	RPM	
		LOAD HP	LOAD HP	
		25.00%	100.00%	
		OIL TEMP (°F)	OIL TEMP (°F)	

TECHNICIAN: _____ SUPERVISOR: _____

B) After the test, grab the patch and the information will be writed down on the format **FO-CRC-09**.

C) The patch will be attached to the format **FO-CRC-09** with HP, RPM and Oil Temperature values during that patch time.



D) The technician in charge of the test must sign the format and send it to the supervisor.

MATCO CAT		PATCH TEST DYNAMOMETER REPORT		CODE
				FO-CRC-09 R0
ORDER NUMBER :	DATE:			
ENGINE NUMBER:	CLIENT:			
SERIAL NUMBER:	REPAIR:			
		0-5 MINUTES	1/2 LOAD	
		RPM	RPM	
		LOAD HP	LOAD HP	
		0.00%	50.00%	
		OIL TEMP (°F)	OIL TEMP (°F)	
		5-10 MINUTES	FULL LOAD	
		RPM	RPM	
		LOAD HP	LOAD HP	
		0.00%	100.00%	
		OIL TEMP (°F)	OIL TEMP (°F)	
		10-15 MINUTES	MAX TORQ	
		RPM	RPM	
		LOAD HP	LOAD HP	
		25.00%	100.00%	
		OIL TEMP (°F)	OIL TEMP (°F)	

TECHNICIAN: _____ SUPERVISOR: _____

2) Uploading the format to the System.

A) Take photos of the patches and upload them to the company's system to keep the record online.

MATCO CAT		Administración de Talleres	
		Taller: Todos, Monitoreo al Sistema (Cerrar sesión)	
Ordenes de Reparación			
Orden de Reparación: 01-02-012722			
Orden Reparación: 01-02-012722		Cliente: 013012 - 9 - MINERA MONTERDE S DE RL DE CV	
Modelo Equipo: 789C - Camión fuera de carretera		Serie: 28W01259	
Codigo Componente: 1000 - Motor		Estatus: CE - Cerrado	
Consultar	Cargos y Descuentos	Segmentar	Cotización de Refacciones
Miscelaneo	Trabajo	Registros de Maquina	Comentarios
Archivos	Autorizaciones	Reporte de Producción	Carta Presupuesto

B) The picture must be clear in each of the tested patches as well as the information of the engines that was captured in the format **FO-CRC-09**

MATCO CAT		Reporte patch test	
Folio:	Orden:		
Ant:	Revisión:		
Motor:	Cliente:		
		0-5 Minutos	10-15 Minutos
		RPM	RPM
		LOAD HP	LOAD HP
		OIL TEMP	OIL TEMP
		FULL LOAD	
		RPM	
		LOAD HP	
		OIL TEMP	

Supervisor: _____ Técnico: _____

CRC-FTPT-V1.0 Última actualización: 05/08/2019

Quality and safety consideration:	Revisión	Fecha
1. Make sure there is no visible leakage in the connections and that the oil tank is not completely full.	00	
2. When the patch is placed on the chamber, verify there is no visible contamination.	01	
3. Verify that all the connections and fixtures are properly tight.	02	
3. When in doubt, ask the supervisor before acting.	03	
4. Wear your personal protective equipment at all times.	04	